Unit 4 Review: Intersection of Lines and Planes

13. Find the intersection, if any, of
    a. the line $\vec{r} = (0, 0, 2) + t(4, 3, 4)$ and the line $\vec{r} = (-4, 1, 0) + u(-4, 1, -2)$
    b. the line $x = t, y = 1 + 2t, z = 3 - t$ and the line $x = -3, y = -6 + 2u, z = 3 - 6u$

15. Find the coordinates of the foot of the perpendicular from $Q(3, 2, 4)$ to the line $\vec{r} = (-6, -7, -3) + t(5, 3, 4)$.

ANSWERS:
13a) $(0,0,2)$ b) no solution
15) $(4,-1,5)$

6. Determine the point of intersection of the two lines
   $(x, y, z) = (-2, 0, -3) + t(5, 1, 3)$ and $\frac{x-5}{-1} = \frac{y-8}{2} = \frac{z+6}{-3}$

7. Let $l_1: x = -8 + t, y = -3 - 2t, z = 8 + 3t$ and $l_2: \frac{x-1}{2} = \frac{y+1}{1} = \frac{z}{3}$ be two lines in three-dimensional space.
   a. Show that l_1 and l_2 are skew lines (that is, neither parallel nor intersecting).
   b. State the coordinates of P_1, the point on l_1 determined by $t = -2$.
   c. Determine the coordinates of P_2, the point on l_2 such that $P_1 P_2$ is perpendicular to l_2.

ANSWERS:
6a) $(8,2,3)$
7b) $P_1(-10,1,2)$ c) $P_2(-1,-2,-3)$

13. A normal to the plane $4x - 2y + 5z - 9 = 0$ passes through the origin. At what point does this normal intersect the plane?

14. Determine where the plane $4x + 5y - z + 20 = 0$ meets the coordinate axes

17. For what values of k will the planes $2x - 6y + 4z + 3 = 0$ and $3x - 9y + 6z + k = 0$
    a. not intersect? b. intersect in a line? c. intersect in a plane?

18. A plane passes through the points (1, 0, 2) and (−1, 1, 0) and is parallel to the vector (−1, 1, 1).
    a. Find the scalar equation of the plane.
    b. Find the equation of the line through the point $Q(0, 3, 3)$ that is perpendicular to the plane.
    c. Find the point at which the perpendicular through Q intersects the plane.

19. Find the equation of the plane that passes through the point (3, 0, −4) and is perpendicular to the line of intersection of the planes $x + 2y - 7z - 3 = 0$ and $x - 5y + 4z - 1 = 0$.

21. Are the two planes $\vec{r} = (4, 0, 3) + t(-8, 1, -9) + u(-1, 5, 7)$ and $\vec{r} = (-14, 12, -1) + p(1, 1, 3) + q(-2, 1, -1)$ parallel, coincident, or neither?

22. Solve each of the following systems of equations. Give a geometrical interpretation of each system and its solution.

a. $x + 5y - 8 = 0$
$5x - 7y + 8 = 0$

b. $2x - 2y + 4z = 5$
$x - y + 2z = 2$

c. $3x + 2y - 4z + 1 = 0$
$2x - y - z + 3 = 0$

d. $x + 2y - 3z = 11$
$2x + y = 7$
$3x + 6y - 8z = 32$

e. $x - y + 3z = 4$
$x + y + 2z = 2$
$3x + y + 7z = 9$

f. $x + 3y + 3z = 8$
$x - y + 3z = 4$
$2x + 6y + 6z = 16$

g. $x + 2y + z = -3$
$x + 7y + 4z = -13$
$2x - y - z = 4$

h. $3x - 3z = 12$
$2x - 2z = 8$
$x - z = 4$

i. $x + y + z = -3$
$x + 2y + 2z = -4$
$2x + 2y + 2z = -5$

ANSWERS:

11. a. $\frac{54}{\sqrt{37}}$ **b.** $\frac{4}{\sqrt{14}}$ **c.** $\frac{2}{\sqrt{5}}$ **d.** $\frac{3}{2\sqrt{30}}$ **12.** $\frac{22}{7}$ **13.** $\left(\frac{4}{5}, \frac{-2}{5}, 1\right)$
14. (−5, 0, 0), (0, −4, 0), (0, 0, 20) **17. a.** $k \neq \frac{9}{2}$
b. will never intersect in a line **c.** $k = \frac{9}{2}$
18. a. $3x + 4y - z - 1 = 0$ **b.** $\vec{r} = (0, 3, 3) + t(3, 4, -1)$
c. $\left(\frac{-12}{13}, \frac{23}{13}, \frac{43}{13}\right)$ **19.** $27x + 11y + 7z - 53 = 0$
20. a. $4x - y + z = 0$ **21.** coincident **22. a.** in R^2 − 2 lines intersect in the point $\left(\frac{1}{2}, \frac{3}{2}\right)$, in R^3 − 2 planes intersecting in the line $\vec{r} = \left(\frac{1}{2}, \frac{3}{2}, t\right)$ **b.** no solution, 2 parallel planes
c. line $\vec{r} = (-1, 1, 0) + t(6, 5, 7)$ **d.** point (2, 3, −1)
e. no solution, triangular prism
f. line $\vec{r} = (5, 1, 0) + t(-3, 0, 1)$
g. line $\vec{r} = (1, -2, 0) + t(-1, 3, -5)$
h. planes coincident with $x - z = 4$ **i.** no solution, 2 planes are parallel and distinct